Satisfied, Not Solved

What a fairness fix actually does to real decisions

Dheirav

Findings summary — August 24, 2026

Update, 23 August. This deck's numbers are from the 18-population stage and are kept as the record of that stage; `06-status-report.pdf` carries the current state. Since this was typeset: 47 populations, a sealed test passed at 9 of 10 on never-measured states, the theory's regime prediction confirmed 27 of 27, and a second sealed campaign showing the tipping point is population-dependent (0.28–0.65). **Further, 25 August:** 57 populations; three adversarial review panels (twenty-four reviews, no rejects) absorbed; a 2022 anomaly traced to the nominal income label sliding under inflation and resolved; and the deployable deterministic form of the mitigation preserves the predicted direction on 10 of 10 populations.

Terminology. Technical names are given in italics the first time they appear, immediately after the plain description. A full glossary is in `00-plain-english.pdf`.

1 The problem

A program decides who gets a mortgage. It approves **79%** of one group of applicants and **57%** of another. That gap is what people mean by algorithmic bias.

The standard fix is to instruct the program: *approve both groups at the same rate* (*demographic parity*). It works — on the standard benchmark dataset the gap between the two rates falls from 0.186 to 0.018, for less than two percentage points of accuracy.

But there are two completely different ways for a program to obey that instruction:

1. Approve *more* people from the group being turned down.
2. Turn down *more* people from the group getting through.

Option 2 makes the numbers equal without helping anybody. It has a name in the ethics literature: *levelling down.*

The score that certifies the fix cannot tell these apart. It measures only whether the two rates match — not how they came to match, and not whether the total number of approvals went up or down.

2 What was already known, and what was not

Known. A 2023 paper by Mittelstadt, Wachter and Russell showed that these fixes mostly do option 2, and argued it is the normal case rather than bad luck. It is in their title: levelling down "by default".

Also known, and I got this wrong at first. That same paper *also* proposes the remedy — their §6 requires that every group receives at least some minimum approval rate. I originally believed my version of this was new. It is not. I found that by reading their paper properly, and corrected my write-up. What is different about mine is narrower: it works without the program ever being told which group an applicant belongs to, and it is tested on data the programs had never seen, across 26 runs from 15 populations — where they report one dataset, scored on the same data they trained on.

Also known: how to measure who paid. A February 2023 paper by Ferry and colleagues, *When Mitigating Bias is Unfair*, sets out a framework for auditing what one of these fixes does to individuals — how many people it moves, in which direction, and what it does to the overall approval rate. Those are the three things counted throughout this work, so the measurement is theirs and I say so. What is mine is the result of running it widely: they audit one model against one baseline, and never ask whether the direction is a property of the data rather than of the method. It is.

Also known, and published five months before I reached it. A March 2026 paper, *Fairness May Backfire: When Leveling-Down Occurs*, proves that when the program cannot see group membership — the setting studied here — the effect is distribution-dependent and can go either way. I found this by running a novelty check after the fact, not before starting. **The conditionality is therefore not my claim.** What that paper does not have is any experiment: zero datasets appear in it.

Also known: the silence of the score itself. That an equalised score does not say which of the two options produced it is reported in the literature as well — a 2023 paper on fairness across combinations of groups observes that levelling down "often goes unnoticed in the overall performance of the model", and a 2023 auditing framework exists because the aggregate score reports neither how many people were affected nor in which direction. Section 1 above is that known problem restated, not a claim of mine.

Not known: *when* it happens, in practice. Their paper never discusses how generous the system is overall. Their only remark near it is that the benchmark dataset is "more than 75% negatively labelled" — said in passing about accuracy cost, never connected to which of the two options you get. They were working in exactly the regime where their conclusion holds, and did not ask whether it would reverse elsewhere.

The nearest miss. A 2024 paper by Goethals, Delaney, Mittelstadt and Russell studies how the cost of fairness changes with the available budget, and calls this "a factor overlooked in previous evaluations". Two of those four authors also wrote the 2023 paper. But in their setup the number of approvals is *fixed in advance* as a budget, so the total cannot move and the effect reported here is invisible by construction. The group best placed to find this looked at the same dimension and did not make the connection.

3 The main finding

Levelling down is not what these fixes do. It is what they do when the system is stingy.

Count what fraction of *everyone* gets approved — both groups together. Call it the **approval rate** (*selection rate*).

- Approval rate below roughly **0.3**: the fix takes approvals away.
- Approval rate above roughly **0.6**: the fix hands approvals out.
- In between, it flips.

3.1 How this was shown

Two datasets always differ in dozens of ways, so comparing them proves nothing. Instead, **one dataset was used and exactly one thing was changed.**

The dataset predicts whether someone earns more than \$50,000. That figure is an arbitrary choice by whoever built the benchmark. Raise it and fewer people qualify; lower it and most do. **Same people, same information about them, same groups — only the finish line moves.** Automated test code verifies that the people, the features and the groups really are identical between runs.

Income cutoff	Approval rate	Change in total approvals	Lost per one gained
\$100,000	0.030	−29.71%	22.03
\$70,000	0.099	−22.05%	2.18
\$50,000	0.252	−2.34%	1.14
\$30,000	0.598	+0.98%	0.83
\$20,000	0.760	+0.80%	0.75
\$10,000	0.890	+0.08%	0.89

The last column is how many people lost an approval for each person who gained one. At the top of the table, 22 people lose out for every one who benefits. At the bottom, more people gain than lose.

3.2 It holds up everywhere it was checked

Check	Result
A second population	Same pattern, more strongly
A different way of grouping people	Four populations, all agree
Fourteen runs, on four new populations, made *after* the prediction was written down	All consistent; correlation +0.901
Real mortgage decisions, where no arbitrary cutoff exists	Lands exactly where predicted
Two loan products in one market, nothing manipulated	Opposite directions, as predicted
Two datasets it was never tuned on	Both placed on the correct side

3.3 What broke when it was tested harder

Since the previous version of this document the finding has been attacked four separate ways, on five populations each rather than one. Two attacks it survived; two found real limits.

Attack	What it asks	Result
A different learner	Is this just a property of straight-line models?	**5 of 5**
A looser or tighter rule	Is it just an artefact of one setting, 25× apart?	**4 of 5**
A different way of reaching the same approval rate	Is it the approval rate, or how hard the task is?	**4 of 5**
A different fairness definition	Does it hold for a rule that never mentions approval rates?	**Failed**

The third attack is the important one. The original evidence changed the approval rate by changing what counted as success — earning over $20,000 against over $100,000. Those are not the same problem: one is harder than the other. So the evidence could never tell apart "the approval rate drives this" from "the difficulty drives this".

The new test holds the task completely fixed and moves only where the model draws its line. Same people, same features, same scores. **The approval rate is what matters**, in four populations of five, and the two methods agree on *where* the switch happens in three of five. That also explains something the previous version could only report as a puzzle: the switch point sits in a different place for mortgages than for income, and it is a genuine difference between populations rather than an artefact of how the earlier evidence was built.

The fourth attack failed, and it is reported as a failure. Under *equalized odds* — a fairness definition concerned with error rates rather than approval rates — the relationship drops to +0.334 where the original definition gives +0.762 on the very same runs. Adding more populations made it *worse*, not better, which is what a near-absent effect does. The claim is therefore narrower than it was: it applies to fairness rules that constrain *how many people are approved*, and not to fairness rules in general.

One population produced no answer at all. Connecticut's approval rates never fall below the switch point, so the constraint barely moves anything there and the arithmetic has nothing to work with. That exposed a flaw in my own method: the original test fixed a bar for the strength of the relationship but never required that anything actually *move*. I have added that requirement and re-checked all twenty run-sets scored under the old test. **Three become "no answer"; none of them had ever passed on noise**, so no conclusion changes — but the hole was real and it was mine.

3.4 Where the switch actually sits

Measured properly — twelve operating points instead of six, and discarding any arm whose model is worse than doing nothing — the switch point is far more stable than I previously reported:

Population	Field	Switch point
COMPAS	criminal justice	0.511
South Carolina	income	0.530
Oregon	income	0.558
Dutch census 2001	occupational status	0.576
Mortgages (one market)	lending	0.66–0.76

Four populations, three fields, two countries, four separate data sources — all between **0.511 and 0.576**. An earlier version of this work called the switch point "population-specific" on the strength of two much coarser measurements. The practical instruction is now **expect about 0.54 and check**, not *measure it from scratch.*

3.5 The result I did not expect: it belongs to one regime, and the theory names it

Every result here uses one family of method. So I ran the obvious alternative — a *post-processing* method, which adjusts decisions after training instead of changing what is trained — across seventeen populations.

The relationship vanished. Correlation -0.024, against $+0.585$ for my own method on the same populations. Taken alone that is the most damaging result in this work.

It is not fragility. The post-processing method **reads the protected attribute when deciding**; mine never does. That distinction is exactly the one the theory paper builds on: it says that when the attribute is available the direction is **determined** — the disadvantaged group gains and the advantaged loses, always — and only when it is unavailable is the direction open.

So there is nothing for a predictor to predict in that regime. And their prediction for it holds in **18 of 18** of my populations, without exception.

What this costs and buys. Every claim I make is now explicitly about the attribute-blind case — which is where nearly every deployed system sits, because setting different thresholds per group is direct discrimination on its face in most jurisdictions. The claim is smaller, and it is bounded on exactly the side a reviewer would have attacked first.

3.6 The obvious objection, answered in advance

Someone will say: this is just the well-known fact that the two *groups* have different underlying rates.

It is not, and the difference matters. That objection is about a gap *between two groups*. This finding is about the *overall level* for everybody together — a different quantity.

The two do move together, unavoidably: if almost nobody or almost everybody is approved, there is no room for a gap between groups. So the analysis **holds the between-group gap fixed and re-checks**, and the relationship gets *stronger*, not weaker ($+0.801 \rightarrow +0.980$ in one population, $+0.964 \rightarrow +0.994$ in another). That check is the core of the argument, not a footnote to it.

4 How this sits against the theory

Checking my measurements against that 2026 paper produced three results.

Their conditions hold on none of my 26 runs. Their theorem is written over the extrema of a ratio that diverges wherever its denominator approaches zero, and every real dataset has such points. The two regions it compares always overlap, by large margins. The conditions are sufficient rather than necessary, and as stated the theorem cannot be applied to data — which is consistent with the paper containing no experiments.

Their direction is nonetheless right. Relaxed from a strict separation to a comparison of the two regions, it predicts correctly in 24 of 26; a median version gets 25. Both misses are the two smallest effects in the set.

And the approval rate is a proxy for their structural quantity, at $\mathbf{r = +0.935}$. The two rules agree with each other on 25 of 26 runs. Theirs requires the joint distribution of the model's scores and group membership; mine requires last year's approval rate.

So their mathematics says *why*, this work says *how to tell*, and the two agree having been arrived at independently — which is better evidence than either alone.

5 Exactly what levelling down is

There is a small piece of arithmetic that makes the whole thing precise, and it appears not to be written down anywhere.

The fix puts both groups on one shared approval rate. Before the fix, the overall rate was just the average of the two groups' rates, weighted by how many people are in each. So:

> The total number of approvals is unchanged **exactly when** the shared rate lands at that size-weighted average.

Levelling down is nothing more mysterious than the program picking a shared rate *below* that point. This needs no assumptions and held to within 0.076 percentage points across all fourteen runs — four fresh populations — that were done only after this was written down.

6 What I could not work out

Why the flip happens. I derived a candidate explanation from theory, wrote the prediction and its pass mark down in advance, and tested it on fourteen runs across four fresh populations, all done afterwards. It passed the test I set — and was then beaten by a far cruder rule ("approval rate above one half? then it's the good kind"). So the derivation added nothing and is reported as a failure.

The lesson is recorded with it: **I set a pass mark but never set a rival to beat**, so passing did not mean what it looked like it meant.

How big the effect is. The approval rate tells you the direction, not the size. One population loses 2.34% where another loses 20.5% at almost the same approval rate. A candidate explanation for that was tested and refuted.

7 Supporting findings

Combinations of groups get missed. A fix applied to one characteristic leaves people at the *intersection* of two (say, Black women) worse off than any group the fix was told about — and worse than the benchmark dataset suggests (9.0× against 13.2×). **This is not my observation.** Kearns et al. (2018) showed the failure mode; Maheshwari et al. (2023) showed that the harm at the intersection is the larger one and that it does not show up in a model's overall performance figures. What I add is the scale — ten populations — and the condition that the effect needs a sizeable minority to appear at all.

The standard explainability tool does not reveal what happened. A widely used method for showing which information a program relied on reports a +151% shift after the fix. Three explanations were proposed and none survived; most of the apparent shift turns out to be an accounting artifact between two near-identical pieces of information. **The transferable lesson: this tool can move enormously without the program's behaviour changing much.**

On small datasets the fix is drowned by chance. Below roughly 2,500 test cases, re-running with a different random split moves the numbers more than the fix itself does.

The recommended alternative is worse. *Minimax group fairness* is proposed in the literature as the answer to levelling down. On mortgage data it destroyed 10% of all approvals, at 46.7 lost per one gained — on the very data where the ordinary fix *created* approvals. Its objective minimises the worst group's *error rate*, and the group with the most errors is not the group being treated worst.

8 Why any of this matters

Published figures for the domains where these fixes are deployed:

Decision	Approval rate	Direction
Résumé screening (applicant to interview, 2024)	0.02–0.03	takes opportunities away
Elite university admission	0.04–0.10	takes opportunities away
University admission, all four-year US institutions	0.66–0.73	hands them out
US mortgage lending (HMDA 2023 aggregate)	∼0.84	hands them out

They span the entire range and sit on opposite sides of the crossover. My earlier claim — that these domains are nearly all stingy, so the harmful case is the default — was asserted and never checked, and it is wrong. The corrected version is stronger:

Two organisations can deploy the identical constraint, in good faith, and get opposite effects on the number of opportunities granted — with identical fairness reports.

That is why the direction cannot be assumed in either direction, and why a rule computable in advance is worth having. It also validates the data used here: the HMDA populations sit at 0.758 and 0.808 against a national aggregate of ∼0.84, so they are representative of the domain rather than an unusually permissive slice.

9 How carefully this was done

Every result is the average of five runs with different random splits. The program is never allowed to see the protected characteristic. All measures are written from their definitions and checked against a standard library on every run.

Where a result could have gone either way, the prediction and its pass mark were written and committed *before* the experiment ran, and the order is provable from the version history.

That discipline is why the following can be reported honestly: **three of my own predictions failed**; two of my own proposed explanations were disproved by my own experiments; and three claims were corrected after the fact — including my claim that the remedy was new, which the literature had got to first.

10 The question I need answered

A theory paper published in March 2026 proves this effect can go either way, and contains no experiments. I have the experiments, a rule computable from a historical approval rate, and the finding that their conditions are satisfied on none of my 26 runs.

Is that an empirical paper worth publishing on its own — or does it now read as a follow-up to someone else's theorem?

Every number here is regenerated from stored results by an automated check that fails if a document and its data disagree. The cited papers are in `papers/`; the full research record is 16 documents in `research/docs/`.